

GROSS ANATOMY OF THE UPPER LIMB

The **Upper limb** is the organ of manual activity and it is divided into the Shoulder, Arm, Forearm, Wrist and the Hand.

I. BONES OF THE UPPER LIMB

This contains the bones of the superior Appendicular Skeleton: The clavicle and scapula (pectoral girdle), Humerus (arm), Ulna and radius (forearm), Carpal bones (wrist), Metacarpals (hand) and Phalanges (fingers).

BONES OF THE PECTORAL GIRDLE

A. THE CLAVICLE (L. little key):

- This bone extends laterally and almost horizontally across the root of the neck.
- It extends from the **manubrium** of the sternum to the **acromion** of the scapula.
- It connects the upper limb to the **axial skeleton** and the **trunk**.
- The **triangular-shaped** medial (sternal) end of the clavicle articulates with the sternum at the **sternoclavicular joint**.
- The medial two-thirds of the **body** (shaft) of the clavicle are convex anteriorly, whereas the lateral one-third is flattened and concave anteriorly.
- Its curvature increases its resilience. The broad lateral (acromial) end of the clavicle articulates with the acromion of the scapula at the **acromioclavicular joint**.
- **Note** : conoid and deltoid tubercles, trapezoid and oblique lines,
 - clavicle is the first bone to begin **ossification**

Functions: To act as a strut for holding the upper limb free from the trunk so it may have maximum freedom of action; to provide attachment for muscles; to transmit forces from the upper limb to the axial skeleton.

B. THE SCAPULA (shoulder blade):

- This flattened, triangular bone lies on the posterolateral aspect of the thorax, covering parts of the **2nd to 7th ribs**.
- The scapula connects the clavicle to the humerus. It is highly mobile and has a **head**, **neck** and **body**. The body is thin and translucent.
- The scapula has a **concave** costal or anterior surface (**subscapular fossa**) and a convex posterior surface from which the **spine of the scapula** projects.
- The smaller part, which is superior to the spine, is called the **supraspinous fossa**, and the larger part, which is inferior to the spine, is called the **infraspinous fossa**.
- The spine continues laterally into a flattened process called the **acromion**. It projects anteriorly and articulates with the **clavicle**.
- **Superolaterally**, the scapula has a shallow **glenoid fossa (cavity)** for articulation with the head of the humerus. This part of the scapula, called the **head**, is connected to its blade-like **body** by a short **neck**.

- The **coracoid process**, like a **bird's beak**, arises from the superior border of the **head** and projects **superoanteriorly**. The **scapular notch** is found in the superior border.
- Locate the **medial** (vertebral), **lateral** (axillary) and superior borders as well as **superior**(T2), **inferior** (T7) and **lateral angles**.
- **Clinically** the arm is measured from the tip of the acromion to the lateral epicondyle of the humerus.

THE BONE OF THE ARM (brachium)

A. THE HUMERUS

- The humerus is the **largest** bone in the upper limb.
- Its smooth, ball-like **head** articulates with the **glenoid fossa** of the scapula.
- Close to the head are the **greater** and **lesser tubercles** for the insertion of the muscles that surround and move the shoulder joint.
- The lesser tubercle is separated from the greater tubercle by the **intertubercular groove** (sulcus), from which lies the tendon of the long head of **biceps brachii** muscle.
- There are also **crest of greater tubercle** and **crest of lesser tubercle**.
- The **anatomical neck** separates the **head** and **tubercles**.
- Distal to the anatomical neck is the **surgical neck**. This is where the bone narrows to become the shaft. The region is called the surgical neck because it is the most frequent **fracture site** of the proximal end of the humerus.
- The **body**, or shaft, of the humerus is easy to palpate, as are its **medial** and **lateral epicondyles**. Its proximal half is cylindrical.
- **Anterolaterally**, there is a roughness known as the **deltoid tuberosity** for the insertion of the deltoid muscle.
- There is a shallow, oblique **radial (spiral) groove** for the **radial nerve** that extends **inferolaterally** on the posterior aspect of the body.
- The distal end of the humerus is expanded from side to side. The **trochlea** (L. pulley) fits into the **trochlear notch** of the ulna, which **swings on** this pulley when the elbow is flexed.
- Just proximal to the trochlea are the **coronoid fossa** (anterior) and the **olecranon fossa** (posterior), which accommodate corresponding parts of the ulna.
- Adjoining the lateral part of the trochlea is a rounded ball of bone called the **capitulum** (L. little head). Proximal to it is **radial fossa**.
- A prominent process, the **medial epicondyle**, projects from the trochlea, and the **lateral epicondyle** projects from the capitulum. The epicondyles being subcutaneous are easily felt. The medial epicondyle is more prominent.
- From each epicondyle, a bony ridge runs proximally; these are known as the **medial** and **lateral supracondylar ridges**, respectively.

BONES OF THE FOREARM (antebrachium)

A. THE RADIUS : This is the **shorter** of the two forearm bones.

- It was given its name because it resembles the **spoke of a wheel**.
- The proximal end of the radius has a **disc-shaped head**, a smooth **cylindrical neck**, and an oval prominence or **tuberosity** (radial tuberosity) distal to the neck.
- The **body** (shaft) of the radius **increases in size** from its proximal to its distal end; it has a slight **lateral convexity** or bowing.
- The body is **concave anteriorly** in its proximal **three-fourths** and **flattened** in its distal one-fourth.
- The **anterior oblique line** of the radius runs obliquely across the body from the region of the radial tuberosity to the area of greatest bowing.
- The medial aspect of the body has a **sharp interosseous border** for attachment of the interosseous membrane. Its **lateral border** is rounded. Note: also **posterior border**.
- The distal end of the radius has a medial **ulnar notch** into which the head of the ulna fits, forming the distal **radioulnar joint**.
- Laterally the distal end of the radius tapers abruptly into a prominent **pyramidal styloid process**.
- The inferior surface of the distal end of the radius is **smooth and concave** where it articulates with the **wrist or carpal bones**.
- Posteriorly there is a prominent **dorsal tubercle** on the distal end of the radius.
- **Note:** grooves for tendons of extensor carpi radialis , extensor digitorum, extensor indicis, extensor pollicis longus and brevis on the posterior part of the radius.

B. THE ULNA (l. elbow): is the longer bone of the forearm.

- This **prismatic** bone looks somewhat like a **pipe wrench**, with the **olecranon** resembling the upper jaw, the **coronoid process** the lower jaw, and the **trochlear notch** the mouth.
- The **proximal "wrench-like" end** of the ulna is larger than the small, rounded **distal end** called the **head**.
- The lateral side of the **coronoid process** has a small, shallow **radial notch** for the disc-shaped head of the radius.
- Inferior to the radial notch is the triangular **supinator fossa**, which provides an attachment for the supinator muscle. This fossa is bounded posteriorly by a distinct **supinator crest**.
- The irregular anterior surface of the **coronoid process** is rough and ends distally in a **tuberosity** onto which the **brachialis**, the chief flexor muscle of the forearm, inserts.
- The **body** (shaft) of the ulna is thick proximally. Its prominent lateral edge, the **interosseous border**, is where the interosseous membrane attaches.

- The small, slender distal end of the ulna has a **rounded head** and a **conical styloid process**. The styloid process projects distally, about 1 cm proximal to the styloid process of the radius.
- The distal end of the ulna has a **convex articular surface** on its lateral side for articulation with the **ulnar notch** of the radius (**distal radioulnar joint**).

BONES OF THE WRIST AND HAND

A. THE CARPUS (L. wrist)

- The eight small bones of the wrist, called **carpal bones** are referred to collectively as the **carpus**. They are arranged in **proximal** and **distal rows**, each containing four bones.
- The proximal row of carpal bones (lateral to medial) consists of the **scaphoid** (navicular), **lunate**, **triquetrum**, and **pisiform**.
- The boat-shaped **scaphoid** is the largest bone of the proximal row and was given its name because of its resemblance to a **rowboat** (G. scaphe). The **lunate** is **moon-shaped**.
- The **pea-shaped pisiform** (L. pisum=pea) is included in the proximal row, even though it is a **sesamoid bone** in the tendon of **flexor carpi ulnaris** muscle. The pisiform bone is a clinically important landmark that is easily palpable.
- The distal row of carpal bones (lateral to medial) consists of the **trapezium**, **trapezoid**, **capitate**, and **hamate**.
- The **hamate** can be identified by its prominent process, the **hook of the hamate**, which projects anteriorly. The **capitate** has a rounded head (L. caput=head) & is the largest of the distal row.

B. THE METACARPUS

- The five metacarpal bones are **miniature long bones**.
- They extend from the **carpus** (wrist) to the **digits** (thumb and fingers) and are numbered from the lateral side. The **first metacarpal** is much shorter than the others.
- Although covered with tendons, the metacarpals can easily be palpated throughout their whole length on the **dorsum of the hand**.
- The **heads** of the metacarpals are at their distal ends, where they articulate with the phalanges. They form **knuckles** of the hand that become visible when the **fist is clenched**.
- On the dorsal surface of each head is a **small tubercle** on each side for attachment of **collateral ligaments** and **joint capsules**.
- The **bodies** (shafts) of the metacarpal are **slightly concave** on their medial and lateral sides, where the **dorsal interosseous muscles** attach.
- The **bases** of the metacarpals are arranged in a fan-shaped manner from the distal row of carpal bones.

C. THE PHALANGES

- Each **phalanx** (bone of a digit) is a **miniature long bone**, which consists of a body (shaft), a larger proximal end or **base**, and a smaller distal end or **head**.
- The **thumb** (first digit) has two phalanges (proximal and distal) and each finger (second to fifth digits) has three phalanges (**proximal, middle and distal**). The phalanges in the first digit are shorter and broader than those in the other digits.
- The **proximal phalanges** are the longest and the distal ones are the shortest.

II. JOINTS OF THE UPPER LIMB

The **pectoral girdle** (clavicle and scapula) connects the upper limb to the trunk; therefore, its articulations are included with those of the upper limb.

A. THE STERNOCLAVICULAR JOINT

Type: saddle type of synovial joint and is the only bony articulation between the upper limb and the axial skeleton (synovial double planar)

Articulation: manubrium of the sternum, medial part of clavicle and medial part of the first costal cartilage.

Cartilages: articular cartilage, articular disc, costal cartilage

Ligaments: anterior and posterior sternoclavicular, interclavicular, costoclavicular

Articular capsule: fibrous and surrounds the entire joint

Movements: Multiaxial (like ball and socket joints)

Stability: is by the ligaments and the articular disc than bones and muscles

AA: from internal thoracic artery axillary a and suprascapular artery (from subclavian a.)

NN: from medial supraclavicular nerve and nerve to subclavius muscle

Weakness: rare. clavicle usually breaks at its medial or lateral thirds before dislocation occurs.

B. THE ACROMIOCLAVICULAR JOINT

Type: slightly movable planar

Articulation: lateral end of clavicle and articular surface of acromion

Cartilages: articular cartilage, articular disc which is fibrocartilagenous and incomplete

Ligaments: acromioclavicular ligament (sup. & inf), coracoclavicular ligament (conoid and trapezoid parts)

Articular capsule: fibrous, encloses the joint and attached to margins of its articular surfaces.

Movements: rotation of acromion on the clavicle (movement anterior and posterior)

Stability: is by coracoclavicular ligament (clavicle always in contact with acromion)

AA: from suprascapular artery and thoracoacromial artery

NN: from supraclavicular, lateral pectoral and axillary nerves

Weakness: "shoulder separation," disruption of acromioclavicular joint and

coracoclavicular ligament (makes the acromion more obvious)

C. THE SHOULDER JOINT(glenohumeral joint)

Type: multiaxial, ball and socket synovial joint, but mobility is gained at the expense of stability

Articulation: spheroidal head of humerus and shallow glenoid fossa

Cartilages: hyaline articular cartilage, glenoid labrum

Ligaments: glenohumeral (superior, middle, inferior), transverse humeral, coracoacromial and other accessory structures coracoacromial arch include long head of biceps, subscapular bursa and rotator cuff muscles.

Articular capsule: thin, loose and fibrous (thus, multiaxial movement is possible) attached to glenoid fossa (medially), coracoid process (superiorly), anatomical neck (laterally).

Movements: has more freedom of movement than any other joint in the body.

Stability: free movement of this joint leads to instability due to the shallowness of the glenoid fossa and the laxity of the articular capsule. Its strength results mainly from muscles surrounding it. Especially, rotator cuff muscles (supraspinatus, infraspinatus, teres minor and subscapularis).

Bursae: subscapular, subacromial

AA: from anterior and posterior circumflex humeral and suprascapular arteries.

NN: from suprascapular, axillary and lateral pectoral nerves.

Weakness: the inferior part lacks tendinous support and this is where it usually dislocates (anterior dislocation is most common). Additionally, rotator cuff wear (supraspinatus) and tear affect it.

D. ELBOW JOINT

Type: uniaxial joint, hinge type of synovial joint

Articulation: trochlea and capitulum of the humerus articulate with the trochlear notch of the ulna (**humero-ulnar**) and the head of the radius (**humero-radial**), respectively.

Cartilages: hyaline articular cartilage

Ligaments: radial collateral, ulnar collateral (anterior, posterior & oblique bands)

Articular capsule: completely encloses the joint but its anterior and posterior parts are thin and weak

Movements: flexion, extension

Stability: jaw-like trochlear notch, spool-shaped trochlea, collateral ligaments

AA: by anastomosis around the elbow which are formed by the collateral branches of brachial artery and radial and ulnar recurrent arteries

NN: is mainly by musculocutaneous, radial, ulnar, median and intrinsic nerves

Bursae: subcutaneous olecranon, subtendinous olecranon, radioulnar, bicipitoradial bursa

Weakness: ulna most commonly dislocated posteriorly

E. PROXIMAL RADIOULNAR JOINT

Common to both: interosseus membrane, oblique cord and quadrate ligament

Type: pivot synovial

Articulation: head of radius and radial notch of ulna

Ligaments: anular & quadrate ligaments

Articular capsule: continuous with the fibrous capsule of the elbow joint

Movements: radius on ulna(135 degrees) i.e pronation and supination

AA: anastomoses around the elbow region

NN: musculocutaneous, median and radial nn

F. DISTAL RADIOULNAR JOINT

Type: pivot synovial

Articulation: head of the ulna and ulnar notch of radius

Cartilage: fibrocartilaginous articular (ulnar) disc

Ligaments: anterior and posterior transverse bands

Articular capsule: fibrous and encloses the joint

Movements: makes possible pronation and supination

AA: anterior and posterior interosseous arteries(ulnar a.)

NN: anterior(median n.) and posterior interosseous nerves (radial n.)

G. WRIST JOINT (radiocarpal)

Type: condyloid, synovial

Articulation: distal radius, articular(ulnar) disc, proximal carpal bones (act as a unit)

Cartilages: ulnar disc, articular cartilages of participating bones

Ligaments: dorsal and palmar radiocarpal, radial and ulnar collateral, interosseous

Articular capsule: fibrous, encloses the joint(strength by the ligaments)

Movements: adduction, abduction, flexion, extension and circumduction are possible

AA: dorsal and palmar carpal arterial arches

NN: anterior interosseous(median), posterior interosseous(radial), dorsal and deep branches of the ulnar nerve

Stability: depends on the ligaments

Weakness: wrist fracture (colles' fracture) i.e shortening of radius

H. INTERCARPAL JOINTS (mid carpal)

Type: plane synovial

Articulation: between proximal and distal rows of carpal bones & individual carpal bones

Cartilages: articular

Ligaments: dorsal and palmar radiocarpal, radial and ulnar collateral, interosseous

Articular capsule: encloses the joint and helps to unite carpal bones

Movements: mainly flexion and abduction

AA: palmar and dorsal arches

NN: same as wrist joint

Weakness: scaphoid fractures & lunate dislocation is common

I: CARPOMETACARPAL AND INTERMETACARPAL JOINTS

Type: plane synovial that permit gliding movement and share common joint cavity with intercarpal joints)

Articulation: distal row carpal bones and combined metacarpals 2-5

Ligaments: dorsal and palmar radiocarpal, radial and ulnar collateral, interosseous
dorsal & volar carpometacarpal, dorsal & volar intermetacarpal ligaments

CMJ of the Thumb

- **Type:** multiaxial, saddle synovial
- **Articulation :** trapezium with base of first metacarpal bone
- **Articular capsule:** fibrous, encloses the cavity and is loosely held
- **Movements:** all angular movements and opposition
- **AA:** dorsal and palmar metacarpal aa, dorsal carpal and deep palmar arterial arches
- **NN:** same as intercarpal joints

J. METACARPOPHALANGEAL JOINTS (MPJ)

Type: condyloid synovial

Articulation: head of metacarpal bones with bases of proximal phalanges

Ligaments: radial and ulnar collateral, palmar & deep transverse metacarpal

Articular capsule: fibrous and strengthened by the ligaments

Movements: all angular movements and circumduction

AA: digital arteries arising from superficial palmar arch

NN: digital nn from ulnar and median nn

K. INTERPHALANGEAL JOINTS (PiP and Dip)

Type: uniaxial, synovial hinge joints

Articulation: head of one phalanx with base of the more distal one

Ligaments: similar with MPJ and reinforced dorsally by extensor expansions

Movements: flexion and extension

AA: from adjacent digital aa

NN: from adjacent digital nn

THE BRACHIAL PLEXUS

- The brachial plexus is a large and very important plexus of nerves.
- It is situated partly in the neck and partly in the axilla.
- The **supraclavicular part** (rami and trunks with their branches) of the brachial plexus is in the posterior triangle of the neck, and its **infraclavicular part** (cords and their branches) is in the axilla.
- The brachial plexus is formed by the **union of the ventral rami of nerves C5 to C8** and the **greater part of T1 ventral ramus** (includes C4 if prefixed and T2 if post fixed).
- The rami that form the brachial plexus are often referred to as the "**roots of the brachial plexus**".
- They lie between the scalenus anterior and scalenus medius muscles.

1. Plan of the Brachial Plexus

- The ventral rami from **C5 and C6** unite to form the **superior (upper) trunk**.
- **C7** continues as the **middle trunk**.
- **C8 and T1** unite to form the **inferior (lower) trunk**.
- Each of the 3 trunks divides into **anterior and posterior** divisions, posterior to the clavicle.
- These divisions are of fundamental significance as the **anterior division supply anterior (flexor) parts** and the **posterior division supply posterior (extensor) parts** of the upper limb.
- The **3 posterior divisions** unite to form the **posterior cord**.
- The **anterior divisions of the superior and middle trunks** unite to form the **lateral cord**.
- The **anterior division of the inferior trunk** continues as the **medial cord**.
- The cords of the brachial plexus bear their name in relation to their position to the axillary artery.
- Each cord of the brachial plexus **divides into two terminal branches**.
- **Lateral cord:** musculocutaneous nerve and lateral root of median nerve.
- **Medial cord:** ulnar nerve and medial root of median nerve.
- **Posterior cord:** axillary and radial nerves.
- The three nerves (musculocutaneous, median, ulnar) are arranged like the limbs of a capital **M**.

2. Branches of the Brachial Plexus

A. Supraclavicular Branches of the Brachial Plexus

- These arise from the rami and trunks of the brachial plexus and are approachable through the posterior triangle of the neck.

1. Dorsal Scapular Nerve

- Is formed at the posterior aspect of the ventral ramus of **C5** with frequent contributions from **C4**.
- It pierces the scalenus medius muscle, and runs deep to the levator scapulae muscle, which it helps to supply.
- It then enters the deep surface of the **rhomboid muscles**, which it supplies.

2. Long Thoracic Nerve

- Is formed from the posterior aspect of the ventral rami of **C5, C6 C7**.
- It passes through the apex of the axilla posterior to the other components of the brachial plexus.
- It supplies the serratus anterior muscle.
- The roots of **C5** and **C6** pierce the scalenus anterior muscle and the root of **C7** passes anterior to this muscle.

3. Nerve to the Subclavius Muscle

- This nerve is from the anterior aspect of the **superior trunk**, from **C5**, with occasional additions from **C4** and **C6**.
- It descends posterior to the clavicle and anterior to the brachial plexus to supply the **subclavius muscle**.

4. Suprascapular Nerve

- This nerve arises from the posterior aspect of the **superior trunk**, fibres from the ventral rami of **C5 and C6** and often **C4** (50% of people).
- It supplies the supraspinatus and infraspinatus muscles and the shoulder joint.
- It passes laterally across the posterior triangle of the neck, superior to the brachial plexus and then through the **scapular notch**.

B. Infraclavicular Branches of the Brachial Plexus

- These arise from the cords of the plexus and are approachable through the axilla.

1. Branches of the Lateral Cord

- The lateral cord has **3 branches**; **one side branch**, the lateral pectoral nerve and **two terminal branches**, the musculocutaneous nerve and lateral root of the median nerve.

A. Lateral Pectoral Nerve

- This nerve pierces the **clavipectoral fascia** to supply the pectoralis major muscle.
- It also sends a branch to the **medial pectoral nerve**, which supplies the pectoralis minor muscle.
- This nerve is so named because it **arises from the lateral cord**; **NOT** because it is lateral to the medial pectoral nerve.

B. Musculocutaneous Nerve

- This nerve supplies the muscles of the anterior aspect of the arm.
- It enters deep to the surface of the coracobrachialis muscle and supplies it.
- It then continues to supply the biceps brachii and the brachialis muscles.
- Just proximal to the elbow joint, the musculocutaneous nerve pierces the deep fascia and becomes superficial.
- From here it is known as the **lateral antebrachial cutaneous nerve** (lateral cutaneous nerve of the forearm) and it supplies the skin on the lateral aspect of the forearm.

C. Lateral Root of the Median Nerve

- This is joined by the medial root of the median nerve, lateral to the axillary artery, to form the median nerve.

2. Branches of the Medial Cord

- The medial cord of the brachial plexus has **5 branches**; three side branches and two terminal branches.

A. Medial Pectoral Nerve

- This enters the deep surface of the pectoralis minor muscle, and supplies it.
- It also supplies part of the pectoralis major muscle.
- This nerve is so named because it **arises from the medial cord**, **NOT** because it is medial to the lateral pectoral nerve.

B. Medial Brachial Cutaneous Nerve

- This nerve supplies the skin over the medial surface of the arm and the proximal part of the forearm.
- It usually communicates with the **intercostobrachial nerve**, which supplies the skin of the floor of the axilla and adjacent areas.

C. Medial Antebrachial Cutaneous Nerve

- This nerve runs between the axillary artery and vein to supply the skin over the medial surface of the forearm.

D. The Ulnar Nerve

- The ulnar nerve passes through the arm into the forearm and hand, where it supplies one and one-half muscles in the forearm, most small muscles in the hand, and some skin.

E. Medial Root of the Median Nerve

- This joins with the lateral root to form the median nerve.
- Median nerve supplies the flexor muscles of the forearm, except flexor carpi ulnaris, and part of the skin of the hand.

3.Branches of the Posterior Cord

- The posterior cord of the brachial plexus has **5 branches**; three side branches and two terminal branches.
- Generally, these nerves supply muscles that **extend the joints of the upper limb**.
- These branches also supply cutaneous nerves of the extensor surface of the upper limb.

A. Upper Subscapular Nerve

- This small nerve supplies the subscapularis muscle.

B.Thoracodorsal Nerve

- This nerve runs between the upper and lower subscapular nerves.
- It travels inferolaterally to supply the latissimus dorsi muscle.

C. Lower Subscapular Nerve

- This nerve passes inferolaterally, deep to the **subscapular artery** and **vein**.
- It gives a branch to the subscapularis muscle and ends by supplying the teres major muscle.

D. Axillary Nerve

- This nerve passes through the **quadrangular space** and then winds around the surgical neck of the humerus.
- It supplies the teres minor and deltoid muscles and ends as upper lateral brachial cutaneous nerve .

E. Radial Nerve

- This nerve provides the major nerve supply to the extensor muscles of the upper limb.
- It also supplies cutaneous sensations to the skin to the extensor region, including the hand.
- As it leaves the axilla, the radial nerve runs posteriorly, inferiorly, and laterally between the **long** and **medial** heads of the triceps brachii muscle and enter the **radial groove** in the humerus.

- The radial nerve gives branches to the triceps brachii, anconeus and brachioradialis muscles, and to the extensor muscles of the forearm.

THE PECTORAL REGION

1. The Pectoral Muscles

- The pectoral region contains four muscles, all of which are attached to the pectoral girdle and are associated with movements of the pectoral girdle and upper limb.

A. The Pectoralis Major Muscle

- This large, thick, fan-shaped muscle covers the superior part of the thorax.
- Its lateral border forms the **anterior axillary fold** and most of the **anterior wall of the axilla**.
- The **fascial sheath** enclosing the pectoralis major muscle is attached at its origin to the clavicle and sternum.
- It leaves the lateral border of this muscle to form the **axillary fascia** in the floor of the axilla.
- The **pectoralis major** and the **deltoid muscles** diverge slightly from each other superiorly and along with the clavicle, form the **deltopectoral triangle** (infraclavicular fossa).
- The **cephalic vein**, occupies the furrow between the deltoid and pectoralis major muscles before it enters the **deltopectoral triangle**.

ORIGIN: **clavicular head**-anterior surface of the medial half of clavicle;

sternocostal head-anterior surface of sternum, superior **six** costal cartilages, and **aponeurosis** of the external oblique muscle.

INSERTION: lateral lip of intertubercular groove of humerus.

INNERVATION: lateral and medial pectoral nerves; **clavicular head** (C5 and C6), **Sternocostal head** (C7, C8, and T1)

ACTION: When both parts of the muscle act together, the pectoralis major **adducts** and **medially rotates** the humerus at the shoulder joint.

--Acting alone, **the clavicular head** helps to **flex the humerus** and from this position the **sternocostal head extends the humerus**.

--When the arm is flexed, the sternocostal head **raises the ribs** during **forced inspiration**.

B. The Pectoralis Minor Muscle

- This triangular muscle lies in the **anterior wall of the axilla**, where it is largely covered by the much larger pectoralis major.
- The **pectoralis minor is the landmark of the axilla**.

- Along with the coracoid process of the scapula, it **forms an arch** deep to which pass the vessels and nerves to the upper limb.
- The pectoralis minor is surrounded by **clavipectoral fascia**, a thin sheet of fibrous tissue that runs from the **clavicle** superiorly to the **axillary fascia** inferiorly.

ORIGIN: ribs 3 to 5 near their costal cartilages.

INSERTION: medial border and superior surface of coracoid process of scapula.

INNERVATION: medial pectoral nerve (C8 and T1)

ACTION: stabilizes scapula and holds it against thoracic wall and rotates scapula & elevates ribs of origin if the shoulder is fixed

C. The Serratus Anterior Muscle

- This is a large, foliate muscle that overlies the lateral portion of the **thorax** and the **intercostal muscles**.
- It was given its name (L. serratus, a saw) because the **saw-toothed appearance** of the fleshy digitations at its proximal attachments.

ORIGIN: external surface of lateral parts of ribs 1 to 8.

INSRTION: anterior surface of medial border of scapula.

INNERVATION: long thoracic nerve (C5, C6, and C7).

ACTION : protracts the scapula and holds or fixes it against the thoracic wall.

----Because it is active during punching, it has been called "**the boxer's muscle**".

---- By fixing the **scapula** to the thorax, it acts as an **anchor** for this bone and permits other muscles to use it as a fixed bone to produce movements of the humerus.

-----Inferior fibres of Serratus anterior help to **raise the glenoid fossa** of the scapula.

- "**winged scapula**": paralysis of serratus anterior due to long thoracic nerve injury.

D. The Subclavius muscle

- This small rounded,triangular muscle lies inferior to the clavicle
- Because of its location ,it serves as a protective between the clavicle and subclavian vessels when this bone is broken

ORIGIN: junction of rib one and its costal cartilage

INSERTION:inferior surface of middle third of clavicle

INNERVATION:nerve to subclavius(C5 and c6)

ACTION: draws the clavicle medially and pulls the point of the shoulder anteriorly

---steadies clavicle during shoulder movements and resists the tendency for the clavicle to dislocate at the sternoclavicular joint

The Back and Shoulder Region

- The following back muscles described are the **extrinsic back muscles**, which attach the upper limb to the axial skeleton. The muscles of the shoulder are divided into three groups:
 - (1) Superficial extrinsic muscles (trapezius and latissimus dorsi);
 - (2) Deep extrinsic muscles (levator scapulae, rhomboids, and serratus anterior); and
 - (3) Intrinsic muscles(deltoid,supraspinous,,infraspinous,teres minor& major, and subscapularis).

A. The Trapezius Muscle

- This is a large, flat, triangular muscle that covers the posterior aspect of the neck and superior half of the trunk.
- It was given its name because the muscles of the two sides form a **trapezium**
- The trapezius muscle attaches the pectoral girdle to the skull and the vertebral column and assists in suspending it.

Medial attachments: medial third of superior nuchal line; external occipital protuberance, ligamentum nuchae, spinous processes of C7 to T12 vertebrae.

Lateral attachments: lateral third of clavicle, acromion and spine of scapula.

Innervation: spinal root of accessory nerve (CN XI) and cervical nn. (C3 and C4).

Action: superior fibres of the trapezius **elevate the scapula**;

its middle fibres **retract the scapula**, and

its inferior fibres **depress the scapula and lower the shoulder**.

- superior and inferior fibres act together in the superior rotation of the scapula.

- The trapezius muscles brace the shoulders by pulling the scapulae posteriorly, hence weakness of these muscles result in **dropping of the shoulders**.

B. The Latissimus Dorsi Muscle

- The name of this muscle is derived from the Latin meaning "**widest of the back**".
- This is large, wide, fan-shaped muscle passes between the trunk and the humerus and acts on the shoulder joint and indirectly on the pectoral girdle.

Medial attachments: spinous processes of the inferior six thoracic vertebrae, thoracolumbar fascia, iliac crest, and inferior 3 or 4 ribs.

Lateral attachments: floor of intertubercular groove of humerus.

Innervation: thoracodorsal n. (C6, C7, and C8).

Action: extends, adducts, and medially rotates the humerus at the shoulder joint.

- When climbing, these muscles raise the trunk. In conjunction with the pectoralis major muscle, the latissimus dorsi muscle raises the trunk to the arm, which occurs when performing **chip-ups**.

C. The Levator Scapulae Muscle

- The superior third of this strap-like muscle lies deep to the **sternocleidomastoid** muscle; the inferior third is deep to the **trapezius**.

Medial attachment: posterior tubercles of transverse processes of C1 or C4 vertebrae.

Lateral attachments: superior part of medial border of scapula.

Innervation: dorsal scapular (C5) and cervical (C3 and C4) nn.

Action: elevates the scapula and helps to tilt its glenoid fossa inferiorly by rotating the scapula. It also helps to retract the scapula and fix it against the trunk and to flex the neck.

D. Rhomboid Muscles

- These two muscles lie deep to the trapezius and are not always distinct from each other
- Rhomboid major is about two times wider than the rhomboid minor
- They appear as parallel bands that pass inferolaterally from vertebrae to the scapulae

Medial attachments-Minor: Ligamentum nuchae and spinous processes of C7 and T1

Major: spinous processes of T2 to T5

Lateral attachments: Medial border of scapula from level of spine to inferior angle

Innervation: Dorsal scapular (C4 and C5)

Action: retract scapula and rotate it to depress its glenoid cavity, help serratus anterior to hold the scapula against the thoracic wall and to fix the scapula during movements of the upper limb. Also are used when forcibly lowering a raised upper limb e.g. when driving a stake with a sledge hammer.

Triangle of auscultation:

Boundaries include **Superomedial:** trapezius

Inferolateral: latissimus dorsi

Superolateral: rhomboidus major

- ❖ When the scapula is drawn anteriorly by folding the arms across the chest and the trunk is flexed, the **auscultatory triangle** enlarges and its lateral side is formed by the medial border of the scapula. Parts of the sixth and seventh ribs and the sixth intercostal space becomes subcutaneous; consequently respiratory sounds may be heard clearly with a stethoscope in this location.

The Scapular Muscles

A. The Deltoid Muscle

- This thick, powerful **shoulder muscle** covers the shoulder joint and forms the rounded contour of the shoulder.
- As its name "deltoid" indicates, it is triangular in outline, i.e., it is **shaped like an inverted Greek letter delta**.
Proximal attachments: lateral third of clavicle, acromion and spine of scapula.
Distal attachments: deltoid tuberosity of humerus.
Innervation: axillary nerve (C5 and C6)
- For descriptive purposes, the deltoid may be divided into three parts: anterior, middle, and posterior. It is capable of acting in part or as a whole.
Action: **anterior part** is a strong **flexor** and **medial rotator** of the humerus.
middle part is the chief **abductor** of the humerus.
posterior part is a strong **extensor and lateral rotator** of the humerus.
- In performing these movements, the deltoid works with other muscles, e.g., the anterior part acts with the pectoralis major and coracobrachialis muscles in flexing the arm.
- The deltoid muscle tends to **stabilise the shoulder joint** and helps to hold the head of the humerus in the glenoid fossa of the scapula during arm movements.

B. The Teres Major Muscle

- The teres (L. round) major is a somewhat flattened rectangular muscle that forms a raised oval area on the dorsum of the scapula, beginning at the inferior angle.
- The teres major and the tendons of the latissimus dorsi form the **posterior axillary fold**.
Proximal attachments: dorsal surface of inferior angle of scapula.
Distal attachments: medial lip of intertubercular groove of humerus.
Innervation: lower subscapular nerve (C6 and C7)
- The teres major muscle **adducts and medially rotates the arm**. It can also help to extend it from the flexed position.

- It is furthermore an important stabiliser of the head of the humerus in the glenoid fossa during the abduction of the arm.

The Rotator Cuff Muscles

C. The Supraspinatus Muscle

- This is a rounded, conical muscle and it lies in the **supraspinous fossa** located superior to the spine of the scapula, deep to the trapezius muscle and the **coracoacromial arch**.
- Its tendon is covered by the deltoid muscle.
Proximal attachment: supraspinous fossa of scapula.
Distal attachment: superior facet on greater tubercle of humerus.
Innervation: suprascapular nerve (C4, **C5**, and C6)
Action: helps to abduct the arm, and helps the other rotator cuff muscles to hold the head of the humerus in the glenoid fossa of the scapula during all movements of the joint (i.e., it helps to stabilise the joint).
- The supraspinatus acts with the deltoid during abduction of the arm and **acts strongly when a heavy weight is carried with the upper limb adducted** (e.g., when carrying a heavy suitcase).
- If the deltoid is paralysed, the supraspinatus can partially abduct the humerus.

D. The Subscapularis Muscle

- This is a thick triangular muscle that lies on the costal surface of the scapula and forms part of the posterior wall of the axilla.
- It crosses the anterior aspect of the shoulder joint on its way to the humerus.
Proximal attachment:: subscapular fossa.
Distal attachment: lesser tubercle of humerus.
Innervation: upper and lower subscapular nn. (C5, **C6**, and C7)
Action: medially rotates the arm and adducts it.

E. The Teres Minor Muscle

- This is an elongated, tapering muscle and is often inseparable from the infraspinous muscle, which lies along its superior border.
Proximal attachment : superior part of lateral border of scapula.
Distal attachment : inferior facet on greater tubercle of humerus.
Innervation: axillary nerve (**C5** and C6)
Action: the teres minor **laterally rotates the arm and assists in its adduction**.
It also helps the other muscles of the rotator cuff to stabilise the shoulder joint.

F. The Infraspinatus Muscle

- This is a triangular muscle that occupies most of the infraspinous fossa.
Proximal attachment : infraspinous fossa of scapula.
Distal attachment: middle facet on greater tubercle of humerus.
Innervation: suprascapular nerve (**C5** and C6)
Action: the infraspinatus muscle **laterally rotates the arm** and helps the other muscles of the rotator cuff to stabilise the shoulder joint.

Anatomical spaces on the back of the shoulder

1. **Quadrangular Space:** is a potential intermuscular with boundaries

Superior: subscapularis in front and teres minor behind and in between these two muscles is the capsule of the shoulder joint

Inferior: teres major **Medial:** long head of triceps **lateral:** surgical neck of humerus

- Axillary nerve and posterior circumflex humeral vessels pass through this space

2. Triangular space:

Superior: teres minor **Inferior:** teres major **Lateral:** long head of triceps

- Circumflex scapular artery passes through this space to the infraspinous fossa

❖ The triangular interval :

Superior: teres major **Medial:** long head of triceps **lateral:** lateral head of triceps

- radial nerve and profunda brachii vessels pass through this interval.

Posterior compartment of the Arm and Forearm

A.The Triceps Brachii Muscle

- Is a large, fusiform muscle and is located in the posterior fascial compartment of the arm.
- It is associated with the small **anconeus** muscle at the elbow.
- As its name "triceps" indicates, it has three heads: long, lateral, and medial.

Proximal attachmens: **long head**-infraglenoid tubercle of scapula;

lateral head-posterior surface of humerus, superior (and lateral) to the radial groove;

medial head-posterior surface of humerus, inferior (and medial) to the radial groove.

Distal attachments : proximal end of olecranon and superior part of posterior surface of ulna.

Innervation: radial nerve (C6, C7, and C8)

Action:it is the main extensor or the forearm.

- Because the long head of the triceps crosses the shoulder joint, it also **aids in extension and adduction of the arm.**
- Its long head also steadies the head of the abducted humerus.

B.The Anconeus Muscle

- Is a small triangular muscle and is on the lateral part of the posterior aspect of the elbow.
- It is usually blended with the triceps and should be considered part of the medial head of this muscle.

Proximal attachment: lateral epicondyle of the humerus.

Distal attachment: lateral surface of olecranon and superior part of posterior surface of ulna.

Innervation: radial nerve (C7, C8, and T1)

Action: helps the triceps to extend the forearm. It also abducts the ulna during pronation of forearm and contracts whenever this joint needs to be stabilised against flexion.

Muscles of the Posterior compartment of the Forearm

- For the purposes of description, the muscles of the posterior of the forearm are divided into the **superficial** and **deep groups**.
- Four muscles in the superficial group of extensor muscles (extensor carpi radialis brevis, extensor digitorum, extensor digiti minimi, and extensor carpi ulnaris) are attached by a flattened **common extensor tendon**.
- It is attached to the **lateral epicondyle of the humerus**, the adjacent fascia, and the lateral supracondylar ridge.
- The extensor tendons are held in place by a strong fibrous band, called the **extensor retinaculum**.
- This retinaculum **prevents bow stringing** of the long extensor tendons when the hand is hyper-extended at the wrist joint.
 - How many synovial sheaths are there under the extensor retinaculum?

1. Superficial Muscles

A. The Brachioradialis Muscle

- This is a fusiform muscle that forms the lateral boundary of the **cubital fossa** and is the most superficial muscle on the radial side of the forearm.

Proximal attachment: proximal two-thirds of lateral supracondylar ridge of humerus.

Distal attachment: lateral surface of distal of end of radius.

Innervation: radial n. (C5, C6, C7).

Action: flexes the forearm, especially when **quick movement** is required or when a weight is lifted during slow flexion of the forearm.
- It is used to give power and speed and acts to its best advantage when the forearm is in the **midprone** position. It is therefore **capable of initiating both pronation and supination**.

B. The Extensor Carpi Radialis Longus Muscle

- This is a fusiform muscle that is partly overlapped by the brachioradialis with which it is often blended.

Proximal attachment: lateral supracondylar ridge of humerus.

Distal attachment: base of 2nd metacarpal bone.

Innervation: radial nerve (C6 and C7).

Action: extends and abducts the hand at the wrist joint.

C. The Extensor Carpi Radialis Brevis Muscle

- This is a fusiform muscle and it is shorter than the extensor carpi radialis longus, which covers it.

Proximal attachment: lateral epicondyle of humerus.

Distal attachment: base of 3rd metacarpal bone.

Innervation: deep branch of radial nerve (C7 and C8).

Action: extends and abducts the hand at the wrist joint.

D. The Extensor Digitorum Muscles

- This is the **principal extensor of the medial four digits** and occupies much of the posterior surface of the forearm.

- It divides into four tendons proximal to the wrist, which passes through a **common synovial sheath**, deep to the extensor retinaculum.
Proximal attachment: lateral epicondyle of the humerus.
Distal attachment: extensor expansions of the medial four digits.
Innervation: posterior interosseous nerve (C7 and C8), a branch of the radial nerve.
Action: extends the proximal phalanges and through its collateral reinforcements, the middle and distal phalanges as well.
- It also helps to extend the hand at the wrist joint after exerting its traction primarily on the digits.

E. The Extensor Digiti Minimi Muscle

- This slip of muscle is a partially **detached part** of the **extensor digitorum**.
- Its tendon runs through a **separate compartment in the extensor retinaculum** and then divides into **2 slips**; the lateral one is joined to the tendon of the extensor digitorum.
Proximal attachment: lateral epicondyle of humerus.
Distal attachment: extensor expansion of the 5th digit.
Innervation: posterior interosseous nerve (C7 and C8), a branch of the radial nerve.
Action: extends the proximal phalanx of the 5th digit at the metacarpophalangeal joint and assists in the extension of the interphalangeal joints.
- It also assists with extension of the hand after exerting its traction primarily on the 5th digit.

F. The Extensor Carpi Ulnaris Muscle

- This is a long fusiform muscle, located on the medial border of the forearm.
- It has **2 heads**.
- Its tendon runs in a groove between the head and styloid process of the ulna, within a special compartment of the extensor retinaculum.
Proximal attachment: lateral epicondyle of humerus and posterior border of the ulna.
Distal attachment: base of 5th metacarpal bone.
Innervation: posterior interosseous nerve (C7 and C8), a branch of the radial nerve.
Action: extends and adducts the hand at the wrist joint.
- Acting with the extensor carpi radialis muscles, it extends the hand.
- Acting with the flexor carpi ulnaris, it adducts the hand.

2. The Deep Extensor Muscles

- The deep extensors of the forearm consist of **3 muscles** that act on the 1st digit or thumb (abductor pollicis longus, extensor pollicis brevis, and extensor pollicis longus), supinator and the extensor indicis.

A. The Supinator Muscle

- This muscle lies deep in the cubital fossa and, along with the brachialis, forms its floor.
- The humeral and ulnar heads of attachment of the supinator muscle envelop the neck and proximal part of the body of the radius.
Proximal attachment: lateral epicondyle of humerus, radial collateral and anular ligaments, supinator fossa and crest of ulna.
Distal attachment: lateral, posterior, and anterior surfaces of proximal third of radius.
Innervation: deep branch of radial nerve.
Action: supinates the forearm by rotating the radius.
- It is the **prime mover** in supination of the forearm and hand.
- The **biceps brachii** assists the supinator in rapid and forceful supination, particularly when resistance is required and the forearm is flexed.

B. The Abductor Pollicis Longus Muscle

- This is a long fusiform abductor of the thumb (L. pollex).
- It lies just distal to the supinator muscle and is closely related to the extensor pollicis brevis muscle.
Proximal attachment: posterior surfaces of the ulna and radius, and the interosseous membrane.
Distal attachment: base of the 1st metacarpal bone.
Innervation: posterior interosseous nerve (C7 and C8).
Action: abducts and extends the thumb at the carpometacarpal joint.
- This muscle acts with the **abductor pollicis brevis** during abduction of the thumb and with the extensor pollicis brevis during extension of this digit.
- Its tendon passes deep to the extensor retinaculum in a **common synovial sheath** with the tendon of the extensor pollicis brevis.

C. The Extensor Pollicis Brevis Muscle

- This is a short fusiform extensor of the thumb that lies distal to the long abductor (abductor pollicis longus), and is partly covered by it.
Proximal attachment: posterior surface of radius and interosseous membrane.
Distal attachment: base of proximal phalanx of thumb.
Innervation: posterior interosseous nerve (C7 and C8).
Action: extends the proximal phalanx of the thumb at the metacarpophalangeal joint.
- It also helps to extend the metacarpal bone of the thumb and to extend and abduct the hand.

D. The Extensor Pollicis Longus Muscle

- This is the long extensor of the thumb and it is larger, and its tendon longer than that of extensor pollicis brevis.
Proximal attachment: posterior surface of middle 1/3 of ulna and interosseous membrane.
Distal attachment: base of distal phalanx of thumb.
Innervation: posterior interosseous nerve (C7 and C8).
Action: extends the distal phalanx of the thumb, and in continued action, it extends the metacarpophalangeal and interphalangeal joints of the thumb.
- It also adducts the extended thumb and rotates it laterally.
- It can also abduct the hand.

E. The Extensor Indicis Muscle

- This is a narrow, elongated muscle that lies medial to and alongside the extensor pollicis longus.
Proximal attachments are: posterior surface of ulna and interosseous membrane.
Distal attachments are: extensor expansion of 2nd digit (index finger).
Innervation: posterior interosseous nerve (C7 and C8).
Action: with the extensor digitorum, it extends the index finger at the proximal interphalangeal joint.
- This puts the index finger into the "**pointing position**".
- The extensor indicis also assists in extension of the hand.

Anterior Compartment of the Arm and Forearm

Muscles of the Arm

A. Biceps Brachii Muscle As its name "biceps" indicates, the proximal attachment of this long fusiform muscle has **two heads** (bi, two + cipital, from L. caput, head).

- The two bellies of the muscle unite just distal to the middle of the arm.
- The biceps is located in the anterior aspect of the arm in the anterior fascial compartment.

Proximal attachments: **short head**-tip of coracoid process of scapula;

long head-supraglenoid tubercle of scapula.

Distal attachments: tuberosity of radius and fascia of forearm via **bicipital aponeurosis**.

Innervation: musculocutaneous nerve (C5 and C6).

Action: when the elbow is extended, the biceps is a simple **flexor of the forearm**.

- It is also a **powerful supinator** when the forearm is flexed and when more power is needed against resistance.
- The biceps barely operates at all during flexion of the prone forearm.
- The tendon of the long head of biceps crosses the head of the humerus within the capsule of the shoulder joint and descends in the **intertubercular groove** of the humerus.
- The **bicipital aponeurosis** is a triangular membranous band that runs from the biceps tendon across the **cubital fossa** and merges with the deep fascia over the flexor muscles in the medial side of the forearm.
- The proximal part of the bicipital aponeurosis can be easily felt where it passes obliquely over the brachial artery and medial nerve.
- This aponeurosis affords protection for these and other structures in the cubital fossa.
- It also helps to lessen the pressure of the biceps tendon on the radial tuberosity during **pronation** and **supination** of the forearm.

B. The Brachialis Muscle

- This is a flattened, fusiform muscle that lies posterior to the biceps brachii.

Proximal attachment: distal half of anterior surface of humerus.

Distal attachments: coronoid process and tuberosity of ulna.

Innervation: musculocutaneous nerve (C5 and C6)

Action: the strong brachialis is the **main flexor of the forearm**. It flexes the forearm in all positions and in slow or quick movements.

- When the forearm is being extended slowly, the brachialis steadies the movement by slowly relaxing.
- It always contracts during flexion of the elbow joint and is primarily responsible for maintaining flexion. Because of this, it is the **workhorse among the flexor muscles** of the elbow joint.

C. The Coracobrachialis Muscle

- This is an elongated, narrow muscle in the superomedial part of the arm and is important mainly as a landmark (e.g., the musculocutaneous nerve pierces it).

Proximal attachment: tip of coracoid process of scapula.

Distal attachment: middle third of medial surface of humerus.

Innervation: musculocutaneous nerve (C5, C6, and C7).

- The coracobrachialis **helps to flex and adduct the arm at the shoulder joint**. It also helps to stabilise this joint.

THE CUBITAL FOSSA

- This is a triangular **hollow area on the anterior surface of the elbow**.
- It is bound superiorly by an imaginary line connecting the epicondyles of the humerus, medially by the pronator teres muscle, and laterally by the brachioradialis muscle.
- The **floor of the cubital fossa** is formed by the brachialis and supinator muscles.
- The **roof of the cubital fossa** is formed by deep fascia and includes the **bicipital aponeurosis**.
- The roof is covered by superficial fascia and skin.

Contents of the Cubital Fossa

- This fossa is an important clinical area because it contains the **biceps tendon**, the **brachial artery** and its **terminal branches** (radial and ulnar arteries), the **brachial veins**, and part of the **median** and **radial nerves**.

Muscles of the Forearm

- These muscles can be divided into **flexor-pronator** and **extensor-supinator groups**.
- The flexor-pronator group arises by a common flexor tendon from the **medial epicondyle** of the humerus; this is referred to as the **common flexor attachment** or origin.
- The extensor supinator group arises by a common extensor tendon from the **lateral epicondyle** of the humerus; this is referred to as the **common extensor attachment** or origin.

Front of Forearm

- The anterior forearm muscles are divided into **3 muscular layers**; a deep layer, intermediate layer and superficial layer.
- A septum of deep fascia separates the deep layer of flexor muscles from the superficial and intermediate layers.
- Most of the flexor tendons are held in place by the **flexor retinaculum**, a thickening of the deep fascia of the forearm.
- This retinaculum prevents bow stringing of the tendons when the flexor muscles contract and also help improve the efficiency of the muscles by changing the direction of force of the tendons.

The Superficial and intermediate layer of Flexors

A. The Pronator Teres Muscle

- This is an elongated and narrow muscle and is a **pronator of the forearm** and a **flexor of the elbow joint**.
- It forms the medial boundary of the cubital fossa and has **two heads of proximal attachment**.
Proximal attachments: medial epicondyle of humerus and coronoid process of ulna.
Distal attachment: middle of lateral surface of radius.
Innervation: median nerve (C6 and C7).
Action: pronates the forearm and flexes it.

- It also assists the pronator quadratus during pronation and is a weak flexor of the forearm.

B. Flexor Carpi Radialis Muscle

- This is a long, fusiform muscle that is located **medial to the pronator teres**.
- In the middle of the forearm its fleshy belly is replaced by a long, flattened tendon that becomes **cord-like** as it approaches the wrist, where it is readily palpable.
Proximal attachment: medial epicondyle of the humerus.
Distal attachments: base of 2nd and 3rd metacarpal bones.
Innervation: median nerve (C6 and C7).
Action: flexes the hand and abducts it.
- To reach its insertion, its long tendon passes through a canal in the lateral part of the flexor retinaculum, and through the **vertical groove in the trapezium**.

C. The Palmaris Longus Muscle

- This is a small fusiform muscle that is absent on one or both sides (usually the left) in about 14% of people, but its actions are not missed.
- It may have two bellies and two tendons.
- When present, the palmaris longus tendon is readily visible and palpable.
- Its long, thin tendon passes superficial to the flexor retinaculum.
Proximal attachment: medial epicondyle of humerus.
Distal attachments : distal half of flexor retinaculum and the palmar aponeurosis.
Innervation: median nerve (C7 and C8).
Action: flexes the hand and tightens the palmar aponeurosis.

D. Flexor Carpi Ulnaris Muscle

- This is the most medial of the superficial flexor muscles in the forearm.
- It has **2 heads of proximal attachment**, between which the ulnar nerve passes distally in the forearm.
Proximal attachments : **humeral head**-medial epicondyle of humerus,
ulnar head-olecranon and posterior border of ulna.
Distal attachments : pisiform bone, hook of hamate bone and the 5th metacarpal bone.
Innervation: ulnar nerve (C7 and C8).
Action: flexes and adducts the hand.
- It also fixes the **pisiform bone** when the hypothenar muscles act.

E. The Flexor Digitorum Superficialis Muscle

- This is the superficial muscle in the forearm and it forms an **intermediate layer** between the superficial and deep groups of forearm muscles.
- The **median nerve** and **ulnar artery** passes deep to the heads of this muscle.
- Near the wrist, the flexor digitorum superficialis gives rise to **four tendons**, which pass deep to the flexor retinaculum.
- The tendons are surrounded by a **common flexor synovial sheath**.
- The superficial pair of tendons passes to the 3rd and 4th digits.
Proximal attachment: **humero-ulnar head**-medial epicondyle of humerus, ulnar collateral ligament, and coronoid process of the ulna,
radial head-superior half of anterior border of radius.
Distal attachment: bodies of the middle phalanges of the medial four digits.
Innervation: median nerve (C7, C8, and T1).
Action: flexes the middle phalanges of the medial four digits by flexing the proximal interphalangeal joints.

- It also flexes the proximal phalanges (i.e., metacarpophalangeal joints) and the wrist joint.

Deep Muscles on the Anterior surface of the Forearm

A. The Flexor Digitorum Profundus Muscle

- This is a long, thick (L. profundus, deep) muscle.
- It is the **only one that can flex** the distal interphalangeal joints of the digits.
- It has an extensive proximal attachment to the ulna and interosseous membrane, and "clothes" the anterior aspect of the ulna.
- The flexor digitorum profundus divides into **4 parts** that ends as four tendons.
- They pass posterior to the tendons of the flexor digitorum superficialis and the flexor reticulum.

Proximal attachment: proximal 3/4 of medial and anterior surfaces of the ulna and interosseous membrane.

Distal attachment: bases of the distal phalanges of medial four digits.

Innervation: medial part-ulnar nerve (C8 and T1), lateral part-median nerve (C8 and T1).

Action: flexes the 2nd to 5th digits after the flexor digitorum superficialis muscle has flexed the middle phalanges.

B. The Flexor Pollicis Longus Muscle

- This is the long flexor of the thumb (L. pollex).
- Lies lateral to the flexor digitorum profundus, where it "clothes" the anterior aspect of the radius.
- Its flat tendon passes deep to the flexor retinaculum, **enveloped in its own synovial sheath** on the lateral side of the common flexor synovial sheath.

Proximal attachment: anterior surface of the radius and adjacent interosseous membrane.

Distal attachment: base of the distal phalanx of the thumb.

Innervation: anterior interosseous nerve from the median nerve (C8 and T1).

Action: flexes the distal phalanx of the thumb and, secondarily, the proximal phalanx and the first metacarpal bone.

- The flexor pollicis longus is the only muscle that **flexes the** interphalangeal joints of the thumb.
- It also flexes the metacarpophalangeal and carpometacarpal joints of the thumb, and it may assist with the flexion of the wrist.

C. The Pronator Quadratus Muscle

- This small muscle is quadrangular and pronates the forearm.
- It cannot be palpated or observed, except in dissections, because it is the deepest muscle in the anterior aspect of the forearm.
- It "clothes" the distal fourth of the radius and ulna and the interosseous membrane between them.
- The pronator quadratus is the only muscle that is attached proximally only to the ulna and is attached distally on the radius.

Proximal attachment: distal fourth of anterior surface of ulna.

Distal attachment: distal fourth of anterior surface of radius.

Innervation: anterior interosseous nerve from median (C8 and T1).

Action: pronates the forearm at the superior and inferior radioulnar joints and is the **prime mover in pronation** of the forearm.

- The pronator quadratus initiates pronation and is **assisted by the** pronator teres when more speed and power are needed.
- The pronator quadratus also helps the interosseous membrane to hold the radius and ulna together, particularly when upward thrusts are transmitted through the wrist.

Muscles of the Hand

- The **intrinsic muscles of the hand** are on the **palmar aspect** and are innervated by branches of the ulnar and median nerves.
- They can be divided into **3 groups**:
 - (1) the thumb or thenar muscles in the thenar compartment;
 - (2) the little finger or hypothenar muscles in the hypothenar compartment; and
 - (3) the lumbrical muscles in the central compartment and the interosseous muscles between the metacarpal bones.
- The long flexor tendons of the **extrinsic muscles of the hand**, arise in the forearm and pass to the digits.
- These are located in the central compartment with the lumbrical muscles.

1. The Thenar Muscles

- These muscles (abductor pollicis brevis, flexor pollicis brevis, and opponens pollicis) produce the **thenar eminence**.
- They are chiefly responsible for the movement known as the **opposition of the thumb**.
- The pressure that the opposed thumb can exert on a fingertip is enhanced by the reinforcing action of the adductor pollicis and flexor pollicis longus muscles.
- The thenar muscles are supplied by the **recurrent branch** of the median nerve.

A. The Abductor Pollicis Brevis Muscle

- This is a thin, short, and relatively broad muscle.
- It forms the anterolateral part of the thenar eminence.

Proximal attachment: flexor retinaculum and the tubercles of the scaphoid and trapezium bones.

Distal attachment : lateral side of the base of the proximal phalanx of the thumb.

Innervation: recurrent branch of median nerve (C8 and T1).

Action: abducts the thumb at the carpometacarpal joint and assists the opponens pollicis muscle during the **early stages of opposition** by rotating the proximal phalanx slightly medially.

B. The Flexor Pollicis Brevis Muscle

- This is a short, wide fusiform muscle.
- It is located medial to the abductor pollicis brevis.

Proximal attachment: flexor retinaculum and tubercle of the trapezium bone.

Distal attachment: lateral side of the base of the proximal phalanx of the thumb.

Innervation: recurrent branch of the median nerve (C8 and T1).

Action: flexes the thumb at the carpometacarpal and metacarpophalangeal joints and **aids in opposition** of the thumb.

C. The Opponens Pollicis Muscle

- This is a quadrangular muscle that lies deep to the abductor pollicis brevis and lateral to the flexor pollicis brevis.
Proximal attachment: flexor retinaculum and the tubercle of the trapezium bone.
Distal attachment: lateral side of the 1st metacarpal bone.
Innervation: recurrent branch of the median nerve (C8 and T1).
Action: opposes the thumb. i.e the tip of the thumb is brought in contact with the palmar surface of the 5th digit. The thumb can also be opposed to other digits.
- **Opposition is the most important movement of the thumb.**
- It involves extension initially, then abduction, flexion, medial rotation, and usually adduction.

D. The Adductor Pollicis Muscle

- This is a fan-shaped muscle and is located in the adductor compartment of the hand.
- It has **2 heads** that are separated by a gap through which the radial artery passes.
Proximal attachment: **oblique head**-bases of the 2nd and 3rd metacarpals, the capitate and adjacent carpal bones,
transverse head-anterior surface of the body of the 3rd metacarpal bone.
Distal attachment: medial side of the base of the proximal phalanx of the thumb.
Innervation: the deep branch of the ulnar nerve (C8 and T1).
Action: adducts the thumb and gives power to the grasp.

2. Hypothenar Muscles

- These muscles deal with the movement of the **5th digit**.
- They lie in the hypothenar compartment of the palm with the 5th metacarpal bone.
- The hypothenar muscles produce the **hypothenar eminence**, or ball of the 5th digit.
- They are all supplied by the **deep branch** of the ulnar nerve.

A. The Abductor Digiti Minimi Muscle

- This is a short, wide, fusiform muscle.
- It is the most superficial of the three muscles forming the hypothenar eminence.
Proximal attachment: pisiform bone.
Distal attachment: medial side of the base of the proximal phalanx of the 5th digit.
Innervation: deep branch of the ulnar nerve (C8 and T1).
Action: abducts the 5th digit and helps to flex the proximal phalanx.

B. The Flexor Digiti Minimi Brevis Muscle

- This is a short, wide, fusiform muscle that is variable in size.
- It lies lateral to the abductor digiti minimi muscle.
Proximal attachments : hook of the hamate bone and the flexor retinaculum.
Distal attachments : medial side of the base of the proximal phalanx of the 5th digit.
Innervation: deep branch of the ulnar nerve (C8 and T1).
Action: flexes the proximal phalanx of the 5th digit at the metacarpophalangeal joint.

C. The Opponens Digiti Minimi Muscle

- This is a quadrangular muscle and it lies deep to the abductor and flexor muscles of the 5th digit.
Proximal attachments : hook of the hamate bone and the flexor retinaculum.
Distal attachment: medial border of the 5th metacarpal bone.
Innervation: deep branch of the ulnar nerve (C8 and T1).
Action: draws the 5th metacarpal bone anterior and rotates it laterally, thereby deepening the hollow of the palm, and bringing the 5th digit into opposition with the thumb.

3. The Lumbrical Muscles

- There are 4 slender lumbrical (L. lumbricus, earthworm) muscles-one for each digit.
Lumbricals 1 and 2
Proximal attachments : lateral two tendons of flexor digitorum profundus.
Distal attachments : lateral sides of extensor expansions of the 2nd and 3rd digits.
Innervation: median nerve (C8 and T1).
Lumbricals 3 and 4
Proximal attachments are: medial three tendons of flexor digitorum profundus.
Distal attachments are: lateral sides of extensor expansions of the 4th and 5th digits.
Innervation: deep branch of the ulnar nerve (C8 and T1).
- The lumbrical muscles **flex the digits** at the metacarpophalangeal joints and **extend** the interphalangeal joints.
- They place the digits into the **writing position**.

4. The Interosseous Muscles

- There are **7 interossei muscles** located between the metacarpal bones.
- They are arranged in **2 layers: 3 palmar and 4 dorsal muscles**.

Dorsal interossei 1 to 4

Proximal attachment: adjacent sides of two metacarpal bones.

Distal attachment: extensor expansions and the bases of the proximal phalanges of the 2nd to 4th digits.

Innervation: deep branch of the ulnar nerve (C8 and T1).

Palmar interossei 1 to 3

Proximal attachment: palmar surfaces of 2nd, 4th and 5th metacarpal bones.

Distal attachment: extensor expansions of digits and the bases of the proximal phalanges of the 2nd, 4th and 5th digits.

Innervation: deep branch of the ulnar nerve (C8 and T1).

- The dorsal interossei **abduct the digits** (**DAB: Dorsal ABduct**).
- The palmar interossei **adduct the digits** (**PAD: Palmar ADduct**).
- They also assist the actions of the lumbrical muscles.
- These are important movements in typing, writing, and playing the piano.

READING ASSIGNMENT

Read about:

- *Name & number of arteries found in each compartment of the arm and forearm.*
- *Each branche of the brachial, ulnar and radial arteries.*
- *The course followed by each nerve, artery and vein inside each compartment.*
- *Formation of deep & superficial palmar arterial arches of the hand & their location.*
- *Arrangement of structures superficial & deep to the flexor & extensor retinacula.*

Clinical Anatomy of the Upper Limb

1. Acromioclavicular separation ('shoulder separation') : is an injury to the acromioclavicular (AC) joint and has three grades of severity.

2. Anterior interosseous nerve entrapment: the deep branch of the median nerve, the anterior interosseous, may be trapped around the elbow following a fracture. The result is weakness in the **flexor pollicis longus** or **flexor profundus** muscles of the index and middle finger, making it impossible to flex the distal phalanx.

3. Arterial puncture at the elbow. the site for the brachial artery is just medial and superior to the bicipital aponeurosis (i.e. the same point at which the blood pressure is taken).

4. Arterial puncture at the wrist. the radial artery is easily palpable just proximal to the wrist joint and lateral to the tendon of **flexor carpi radialis** while the ulnar artery can be palpated just proximal to **Guyon's canal** medially.

5. Biceps tendon reflex: is elicited by tapping with a hammer over the dorsal surface of the thumb or forefinger held against the biceps tendon. The reflex (mainly through

C6) causes the biceps muscle to contract and the elbow to flex.

7. Bicipital tendinitis: the tendon of the long head of biceps in the bicipital groove may become frayed and rupture producing a noticeable bulge on the anterior surface of the arm when the elbow is flexed as the distal belly retracts.

8. Carpal tunnel syndrome: is a constellation of symptoms due to compression of the median nerve in the carpal tunnel beneath the **flexor retinaculum**. The sensory loss is usually in the radial two and a half fingers and the thumb, and the presenting symptom may be pain along the radial side of the hand. If the condition is long standing there may be a noticeable loss of muscle bulk in the **thenar eminence**.

9. Colles' fracture: is named after Abraham Colles, a Dublin surgeon. This fracture of the lower end of the radius with posterior displacement is commonly due to a fall on to an outstretched hand. Complications include median nerve irritation, rupture of the extensor pollicis longus tendon, and subluxation of the distal radio-ulnar joint. A **Smith's fracture** is a volarly angulated Colles' fracture.

10. De Quervain's disease: is a chronic inflammatory thickening of the common tendon sheath of the abductor pollicis longus and extensor pollicis brevis muscles as they run across the lower end of the radius near the radial styloid. There is a palpable thickening of the tendon sheath and pain on movements of the thumb.

11. Dislocations of the elbow: while allowing a wide range of motion, the joint has an inherent stability and requires a considerable force to dislocate. For this reason, approximately one-third of elbow dislocations are associated with fractures of the elbow. Most commonly, the elbow dislocates posteriorly.

12. Dislocation of the finger: accidents can cause a 'jamming' force to be applied to the end of the finger, or the finger may be forcefully overextended. Either of these situations, or a combination of both, can result in a dislocation.

13. Dislocation of the humerus: anterior dislocations account for 97% of shoulder dislocations. They are usually due to indirect force, such as from abduction and internal rotation with the humeral head lying inferomedial to the glenoid .

14. Dislocation of the lunate: the lunate lies in the middle of the proximal carpal row . Severe forced dorsiflexion of the wrist may occasionally cause the lunate to become dislocated . When the wrist then returns to the neutral position the lunate is displaced anteriorly, causing pressure on the median nerve.

15. Dislocation of the radial head: seen in young children usually before the age of 5 (also known as a '**pulled elbow**'), is due to subluxation of the radial head from the annular ligament.

16. Dupuytren's contracture: is an hereditary deformity of the hand due to thickening of the palmar aponeurosis with resultant fibrosis and eventual contracture of the fingers .

17. Erb's palsy (Erb–Duchenne palsy): is a brachial plexus injury to the upper roots, often an obstetric injury during a difficult delivery when traction is applied to the baby's head when it is in lateral flexion and the shoulder of the opposite side is pulled downwards. This most common of obstetric brachial plexus injuries involves the fifth and sixth cervical nerve roots.

18. Gamekeeper's thumb: is caused by a valgus force directed on the thumb metacarpophalangeal (MCP) joint to produce a failure of the ulnar collateral ligament resulting in instability accompanied by pain and weakness of the pinch grasp.

19. Golfer's elbow: similar to tennis elbow, this relatively uncommon problem presents as pain over the medial epicondyle and is aggravated by extension of the elbow in a supinated forearm. It is due to repetitive strain from any of the common flexor origin muscles. Steroid injection can give good symptomatic relief.

20. Klumpke's paralysis: is a brachial plexus lesion most commonly due to an abduction strain at the time of childbirth causing injury to the C8 and T1 roots of the brachial plexus, resulting in a **claw hand deformity** due to the paralysis of the intrinsic hand muscles.

21. Mallet finger: is avulsion of the extensor attachment of the extensor digitorum where it inserts into the dorsal terminal phalanx of the fingers. This is often caused by stubbing the finger against a hard object.

22. Median nerve paralysis (palsy): either due to higher interruption from nerve block at the elbow; supracondylar fracture or at the wrist, results in compromise of normal intrinsic motor function of the thumb.

23. Olecranon bursitis: is inflammation of the bursa overlying the olecranon process of the ulna associated with prolonged pressure at this point.

24. Posterior interosseous nerve entrapment: occurs where this branch of the radial nerve passes through two planes of fibres within the supinator muscle, often following elbow trauma or a fibrous band within the supinator.

25.Radial nerve paralysis (palsy): commonly known as a ‘crutch’ palsy or ‘Saturday night’ palsy, is usually caused by compression of the radial nerve in the axilla (crutches fitting poorly into the armpit or a drunken stupor in which arms are flopped over an armchair are common causes). Both elbow and wrist extension is compromised when damage occurs in the axilla.

26. Tennis elbow: tenderness over the common extensor origin on the lateral epicondyle at the elbow, is caused by any repetitive movement (tennis serves, computer entries) which involves the extensor group of muscles.

27.Triceps tendon reflex: elicited by tapping with a hammer the triceps tendon just above the olecranon, extends the elbow. This pathway is mainly through C7. Diminished or exaggerated reflex responses indicate abnormality of function. A cut nerve, for example, will give no reflex.

28. Ulnar nerve paralysis (palsy):commonly results from damage to the nerve behind the medial epicondyle of the humerus. This impairs the power of wrist adduction and causes an inability to spread the fingers and may result in a **claw hand** due to paralysis of the interossei in distal lesions. There may also be an inability to adduct the thumb.

29.Venepuncture of the upper limb: is a very common procedure carried out on most hospital admissions. The most common sites are the **median cubital vein** in the antecubital fossa and the **cephalic vein** just proximal to the wrist joint. In patients with difficult venous access, the dorsal venous arch of the hand may also be used.

30.Venous cutdown: Occasionally the upper limb is used in an emergency as a site for venous cutdown when the long **saphenous vein** of the lower limb is not available (e.g. after vein harvesting. Common sites for cutdown are the cephalic vein and brachial vein.

31. Volkmann's contracture: is a deformity of the upper limb due to muscular ischaemia following injury to the brachial artery typically in a supracondylar fracture of the humerus. An interruption of the blood supply of the upper limb musculature leads to necrosis and eventual fibrosis. Clinically, the finger flexors are usually the most severely affected, the patient being unable to extend the fingers when the wrist is flexed.

32. Winging of the scapula: a scapula that sticks out from the posterior chest like an **angel's wing**, is noticeable when the hand is pushed against a wall or used to open a door. It is most commonly caused by a weakness of the **serratus anterior** or sometimes **latissimusdorsi** muscles. This may be due to muscular disease or nerve palsy.

34. Wrist drop: is a complication of radial nerve palsy due to loss of the forearm extensor muscles and may be associated with a loss of sensation over the first dorsal web space between the **thumb** and **index finger**. There is often a weakness in power grip due to loss of synergistic action of the extensors.